
DSC 245 Final Report: Finding Causal Relationships in Financial Markets

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1 Introduction

The stock market is one of the most studied systems in the world, yet association is constantly mistaken for causation. Correlations between macroeconomic indicators and stock returns are well documented, VIX spikes and markets fall, oil prices rise and the index follows, inflation climbs and the Fed reacts. But knowing that two variables move together tells us nothing about what drives what, what would happen under intervention, or whether both are simply responding to a third variable we have not measured. This distinction, between association and causation, is the central problem motivating this project.

Standard approaches to studying financial markets, including regression analysis, VAR models, and correlation-based portfolio construction, are fundamentally associational. They describe patterns in the data but cannot answer causal questions: does inflation cause the Fed to raise rates, or does the market price in rate changes before the Fed acts? Does Consumer Discretionary lead the S&P 500, or do they simply share exposure to the same business cycle shocks? Did tech stocks cause the S&P 500 to shift toward tech dominance, or did hidden macro forces reward tech and non-tech simultaneously? These questions require a different framework entirely.

This project applies causal discovery and causal inference methods to the S&P 500 with one unified goal, to move beyond correlation and identify the actual causal structure of how the market works. We do this across three connected scales, each representing a different level of the same system.

At the macro level, we ask which economic indicators (among inflation, unemployment, interest rates, oil prices, market volatility, money supply, consumer sentiment, and the 10-year Treasury yield) actually cause S&P 500 returns, and whether that causal structure has changed across monetary policy regimes over 25 years. The rolling correlation analysis motivating this section shows that relationships between macro variables and market returns are highly unstable over time, suggesting that a single static model is insufficient and that regime-aware causal discovery is necessary.

At the sector level, we ask which industries systematically lead the S&P 500 with a time delay, and whether the market in turn drives sectors back. We analyze nine sector ETFs at daily, weekly, and monthly resolution, using multiple methods to distinguish genuine predictive precedence from shared

exposure to common macro shocks. The goal is to identify which sectors, if any, consistently move before the broader market in a way that survives conditioning on all other sectors simultaneously.

At the index composition level, we ask whether the dramatic shift toward tech dominance in the S&P 500 top-10 holdings, from GE and Exxon in 2000 to NVIDIA, Microsoft, and Apple comprising over 20% of the index by 2025, was caused by tech stocks driving the market, or whether something hidden caused both simultaneously. This section uses five monthly time series spanning 310 observations and six causal methods to test whether tech and non-tech returns have any direct causal relationship, or whether the structural shift in index composition is better explained by hidden macro forces operating on both sectors at once.

These three questions are not independent. They are three lenses on the same system examined at different scales. Macro forces set the causal regime within which sectors operate. Sector dynamics shape which companies grow fastest. And the growth of those companies mechanically increases their weight in a cap-weighted index over time. A complete causal picture of the S&P 500 requires all three levels.

To answer these questions rigorously we address the core challenges of financial time series directly. Every variable is tested for stationarity using the Augmented Dickey-Fuller (ADF) test and transformed before any causal modeling. Lagged relationships and autocorrelation are handled through methods specifically designed for time series, including PCMCI and VAR-based Granger causality. Hidden confounders are addressed through FCI, which allows for latent variables rather than assuming causal sufficiency. We are careful throughout to distinguish predictive temporal precedence, which Granger causality measures, from structural causation, flagging where hidden confounders prevent clean causal claims.

Across the three sections we use six causal methods:

1. Granger causality for temporal precedence testing.
2. The PC algorithm for constraint-based causal skeleton discovery.
3. FCI for causal discovery under potential hidden confounding.
4. CD-NOD for detecting regime changes in causal mechanisms.
5. PCMCI for time-series-aware causal discovery that controls for autocorrelation.
6. The backdoor criterion for estimating causal effect sizes from the discovered graph.

Each method was chosen because it addresses a specific aspect of the causal question at hand, and the consistency of findings across multiple independent methods is treated throughout as the primary standard of evidence.

2 Macroeconomic Factors

2.1 Introduction

To get an understanding of how the market moves, an important first scale to look at is the macroeconomic conditions that the market exists in as that is the structural basis for the market itself. Therefore, to get a better understanding, that is the first lens under which we will analyze the stock market.

2.2 Factors Considered

We obviously had to select a subset of all known macroeconomic factors. The factors we selected are diverse, from unemployment to the price of oil, they were picked to cover a broad range of known important market conditions. All of these factors are time varying and not guaranteed to be stationary, which is a common assumption among the techniques we will be applying. Therefore, we had to perform specific transformations to each variable to make it stationary. After performing the transformation all variables were confirmed to be stationary using an Augmented Dickey Fuller Test with $\alpha = 0.05$ [Dickey and Fuller, 1979]. A summary of the factors selected can be found in Table 1. There we also describe the transformations performed and our reasoning for performing the transformation.

Table 1: Macroeconomic factors considered and stationarity transformations performed

Variable	Description	Transformation	Reasoning
S&P 500	A stock market index tracking the top 500 performing companies listed in stock markets in the USA	Monthly log return	Price level is non-stationary
VIX	A measure of the expected volatility of the stock market. It is commonly referred to as the fear index.	Log level	Already mean-reverting
Fed Funds Rate	The target interest rate that banks can lend money to other institutions at.	First difference (MoM)	Level is non-stationary
CPI	An estimate of the level prices of various consumer goods, tracks changes in prices over time.	Year-over-year percent change	Captures inflation rate
Unemployment	The unemployment rate represents the number of unemployed as a percentage of the labor force.	First difference (MoM)	Level is non-stationary
M2	Estimates the total volume of liquid assets in the US economy.	Year-over-year percent change	Captures growth rate
Consumer Sentiment	A consumer confidence index, that assesses how consumers in the US feel about the near and long term future of the economy.	First difference (MoM)	Level is non-stationary
Oil	Tracks the price of crude oil in the US.	Monthly log return	Price level is non-stationary
10Y Treasury	The interest rate that the US government pays to borrow money for 10 years, which acts as a baseline for other money lenders.	First difference (MoM)	Level is non-stationary

2.3 Exploratory Analysis

Before beginning to investigate the causal relationships we wanted to get an understanding of the data and the various correlational relationships. As well as getting an understanding of how these variables and relationships change over time. To this end we generated a few different visualizations such as Figure 14 in the appendix which shows all variables plotted as a time series with specific monetary regimes highlighted. As seen in 15 we also visualized the rolling correlation of each variable with the S&P 500 return to see how the correlational relationships have changed over time. The high variability in these relationships motivates us further to look into causal relationships.

2.4 PC Algorithm

The PC algorithm is a constraint based method of causal discovery that works by exploiting conditional independence relationships to infer the causal graph Spirtes and Glymour [1991]. We chose to use the PC algorithm because of its ease of interpretability and implementation using the causal-learn library Zheng et al. [2024]. Using the PC algorithm we were able to visualize the causal relationships between the various macroeconomic factors and the stock market, we also wanted to understand how these relationships evolved over time so we also did the analysis on three subsets of the data to get a snapshot of what the causal graph looked like underneath different economic regimes.

2.4.1 PC Algorithm Full Sample (1/1/2000 – 12/31/2024)

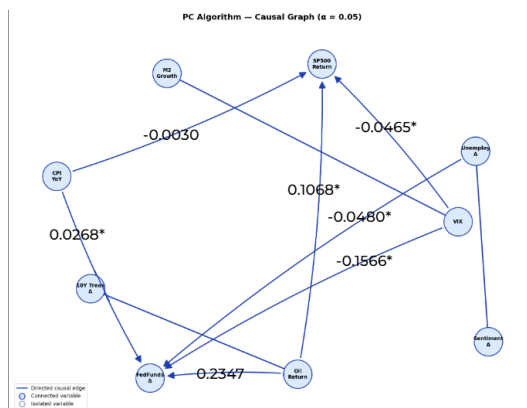


Figure 1: Causal Graph Full Sample (1/1/2000 – 12/31/2024)

We first conducted PC Algorithm on full-sample with effect sizes estimated using the backdoor criterion which can be seen in Figure 1. The effect size estimation with star sign means the effect is significant.

There are two causal pathways identified, the first being the market drivers. We see that VIX and Oil directly cause S&P 500 returns. VIX is commonly called the fear index, and represents the perceived volatility, it makes sense then that it depresses the return. On the other hand, Oil/energy prices, which can be seen as a proxy for global market demand, boost the return. The other causal pathway observed is the Fed reaction function. We see that CPI, Unemployment, and VIX directly drive Fed rate decisions. This finding is consistent with the Taylor principle which describes a formula for setting interest rates Taylor [1993].

We also observe that CPI does not directly move markets, in fact it appears that inflation affects stocks only indirectly through the Fed channel.

2.4.2 PC Algorithm By Regime (1/1/2000 – 12/31/2007)

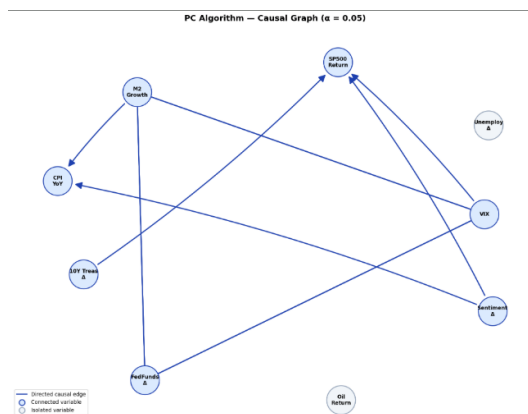


Figure 2: Causal Graph By Regime (1/1/2000 – 12/31/2007)

The first regime we will analyze is from 2000–2007 which is the earliest part of the dataset ending directly before the Great Financial Crisis (GFC). In this regime money supply (M2) is the central node. It drives Fed Funds, inflation, and CPI simultaneously, this is because of Greenspan’s conventional monetary policy. The Fed under Greenspan operated through traditional interest rate channels. The federal funds rate was the primary tool, and M2 money supply was actively monitored as an intermediate target for monetary policy, though it had been downgraded from being a central target. The Fed published M2 targets and explicitly referenced monetary aggregates in policy decisions Greenspan [1993].

We can also see that 10Y Treasury rates, Consumer Sentiment, and VIX directly cause S&P 500 returns, it is possible that sentiment reflected consumer spending power, and VIX captured dot-com and credit crisis fear.

We also know oil price shocks were deliberately excluded from core inflation measures, keeping oil causally disconnected from financial markets. Pre-2007, oil price shocks were treated as temporary supply disruptions. The Fed explicitly looked through oil price volatility ("core inflation" excluding food and energy was the preferred measure). So oil didn’t causally connect to Fed decisions or financial markets in a systematic way Cavallo [2008].

2.4.3 PC Algorithm By Regime (1/1/2008 – 12/31/2019)

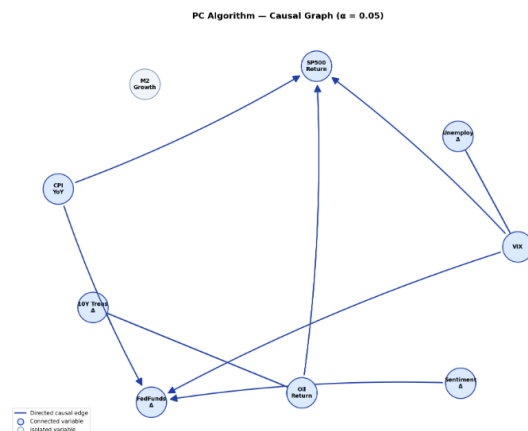


Figure 3: Causal Graph By Regime (1/1/2008 – 12/31/2019)

The next regime we analyzed was from 2008 – 2019 which covers the Great Recession until the Covid-19 Pandemic. In this regime M2 completely disconnects and Oil emerges as the new central driver of both markets and Fed policy. In this regime CPI, Oil, and VIX are the direct drivers of the S&P500 return.

We can see that M2 completely disconnects, because The Fed’s Zero Interest Rate Policy and Quantitative Easing broke the traditional M2 transmission mechanism. Oil emerges as the new central driver of both markets and Fed policy, as the Fed explicitly cited energy prices in rate decisions Federal Open Market Committee [2016].

CPI, Oil Return and VIX are the direct drivers of SP500 Return — CPI signaled Fed tapering, oil reflected global demand and deflation risk, and VIX captured financial stress under unconventional monetary policy.

2.4.4 PC Algorithm By Regime (1/1/2020 – 12/31/2024)

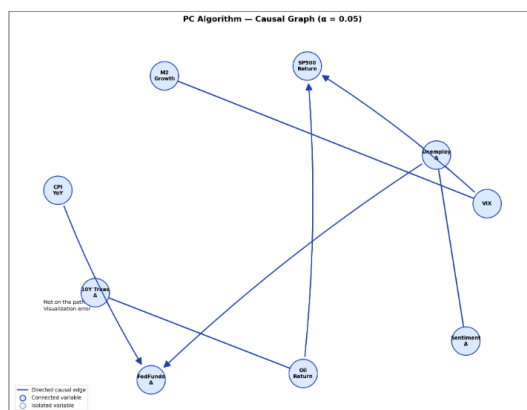


Figure 4: Causal Graph By Regime (1/1/2020 - 12/31/2024)

The final regime we will analyze is from 2020 - 2024 which covers from end of the pandemic until nearly the present day. In this regime the S&P 500 returns are driven by VIX and Oil, not monetary aggregates. We can also see the Fed’s dual mandate operates in both CPI and unemployment directly cause Fed rate decisions

Unlike uantative easing, post-COVID fiscal stimulus injected money directly into households, making market sentiment (VIX) and commodity prices (Oil) the primary transmission channels to equity markets rather than monetary aggregatesMao et al. [2024]. We also observe, the 1977 Federal Reserve Act dual mandate — maximum employment and price stability — operated simultaneously for the first time since the 1980s, as 40-year high inflation and historically tight labor market both independently triggered rate decisions. with CPI and Unemployment directly and independently causing rate decisions for the first time since the 1980s stagflation era Cook [2024].

2.5 Conclusion

Overall, we have observed that the causal relationships between the market and macroeconomic indicators are not stable. They are a product of the economic policy and global economic conditions that exist in a given regime. By partitioning the data and applying our causal discovery techniques to each separately, we are able to recover these conditions and the economic policy that created them.

While the PC algorithm provides a great tool for this further work could include moving towards using FCI or CD-NOD in order to lessen the necessary assumptions that we had to make in order to do our analysis. Also if we want to look more closely than the macroeconomic level it makes sense to take a magnifying lens to the specific industries and see how they relate to each other.

3 Industries

3.1 Introduction

In this section, we focus on industry sector ETFs and their relationship with the S&P 500. Sector ETFs are useful because they represent different parts of the economy, such as Technology, Energy, Healthcare, Financials, and Consumer Discretionary. These sectors may react differently to changes in consumer demand, interest rates, growth expectations, or market risk appetite.

The goal of this section is to study whether some sectors tend to move before the broader market, whether the market itself leads individual sectors, and whether there is a first-mover sector in the broader industry network. We do not interpret causality as a strict structural claim. Instead, we use causality in the limited Granger-style sense: a sector is considered causal-leading if its past returns improve prediction of another sector or the S&P 500 after conditioning on the rest of the observed system.

3.2 Data

The dataset contains log returns for nine sector ETFs and the S&P 500 index. The sector ETFs included are Financials (XLF), Materials (XLB), Energy (XLE), Industrials (XLI), Technology (XLK), Healthcare (XLV), Consumer Discretionary (XLY), Consumer Staples (XLP), and Utilities (XLU). The market index is represented by the S&P 500, denoted as $\hat{GSPC}$.

We analyze the data at three temporal resolutions: daily, weekly, and monthly. The daily dataset contains 6,882 observations, the weekly dataset contains 1,429 observations, and the monthly dataset contains 330 observations. Before running the causal analysis, raw prices were converted into log returns because stock prices are generally nonstationary, while returns are more appropriate for time-series modeling. Augmented Dickey-Fuller tests confirmed that all return series were stationary at the 5% significance level.

3.3 Methodology

We use three complementary methods to evaluate directed temporal relationships between sectors and the S&P 500.

First, we use Conditional VAR Granger causality. This method tests whether lagged values of one variable help predict another variable after controlling for all other variables in the system. This is stronger than a simple bivariate Granger test because it reduces the risk that an observed relationship is actually driven by another sector or by the market index.

Second, we use PCMCI, a causal discovery method designed for time-series data. PCMCI identifies lagged relationships that remain significant after conditioning on other relevant variables. We use PCMCI as a second conditional statistical method because agreement between Conditional VAR and PCMCI provides stronger evidence than either method alone.

Third, we include XGBoost lagged-feature importance as a predictive robustness check. XGBoost is not treated as a causal method. Instead, it helps us check whether lagged sector returns have practical predictive value in a flexible nonlinear model. If XGBoost supports an edge that is also supported by Conditional VAR or PCMCI, we treat that as stronger evidence. If XGBoost is the only method supporting an edge, we interpret it only as predictive signal rather than causal-leading evidence.

For each directed relationship $X \rightarrow Y$, we compute a total conditional score:

$$\text{Total Conditional Score} = \text{CVAR} + \text{PCMCI} + \text{XGB}$$

where each method contributes 1 if it supports the edge and 0 otherwise. We also compute a more conservative core causal score:

$$\text{Core Causal Score} = \text{CVAR} + \text{PCMCI}$$

The core causal score excludes XGBoost because XGBoost is used only as predictive supporting evidence.

3.4 Results

The results show that daily data contains the richest directed structure. This is expected because financial lead-lag effects are often short-lived and may disappear when returns are aggregated to weekly or monthly levels. At the daily resolution, Conditional VAR identifies 21 directed edges, PCMCI identifies 29 directed edges, and XGBoost selects 30 top predictive edges. At the weekly level, the structure becomes weaker, and at the monthly level, Conditional VAR finds no significant directed edges.

The strongest consensus edges are those supported by all three methods. Several of the strongest edges involve Consumer Discretionary (XLY) as the source sector. In particular, XLY leads the S&P 500, Technology, Energy, Consumer Staples, and Utilities in the strongest consensus results. This suggests that XLY occupies a central position in the sector network and may contain early information about broader market conditions.

For the first research question, we find evidence in both directions. Some sectors predict the S&P 500, especially at the daily level. The strongest sector-to-market result is $XLY \rightarrow \hat{GSPC}$, which is supported by Conditional VAR, PCMCI, and XGBoost. Technology also leads the S&P 500, with support from both Conditional VAR and PCMCI. However, the market also predicts several sectors, including Energy, Materials, Technology, Consumer Staples, and Financials. Therefore, the relationship is not one-directional. Instead, the evidence suggests a feedback structure where some sectors contain early market information, while the broad index also transmits information back to individual sectors.

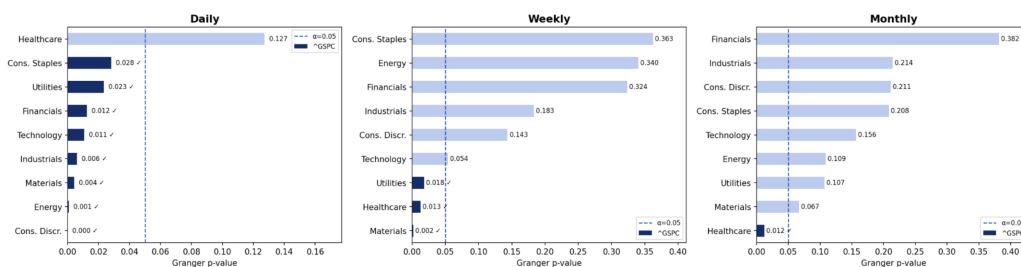


Figure 5: Sectors that Granger Causes GSPC

For the second research question, Consumer Discretionary is the clearest first-mover candidate. XLY has the highest total conditional score and the highest core causal score among all sector source nodes. However, we interpret this carefully. The result does not mean Consumer Discretionary structurally causes the entire market to move. Rather, XLY shows the strongest and most consistent predictive temporal precedence in the observed system. Economically, this is plausible because Consumer Discretionary companies are sensitive to consumer confidence, household spending expectations, credit conditions, and general optimism about the business cycle.

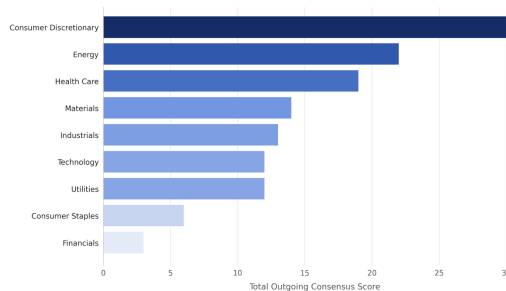


Figure 6: First-Mover Sector Ranking Across All Directed Edges

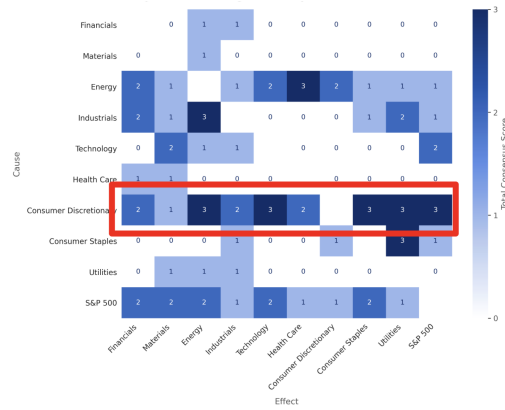


Figure 7: Daily Consensus Edge Heatmap

For the third research question, the strongest sectors leading the S&P 500 are Consumer Discretionary, Technology, and Healthcare. Consumer Discretionary ranks first because the daily edge $XL_Y \rightarrow \hat{GSPC}$ is supported by all three methods. Technology ranks second because $XL_K \rightarrow \hat{GSPC}$ is supported by both conditional statistical methods. Healthcare ranks third, although its evidence is weaker and comes mainly from PCMC and XGBoost. Energy and Consumer Staples receive some predictive support from XGBoost, but they do not have strong core causal support. Financials and Utilities show no support for leading the S&P 500 under our methods.

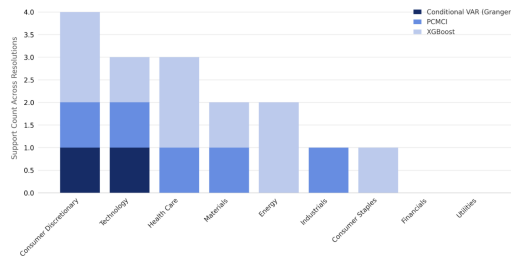


Figure 8: Method Support for Sectors Leading the GSPC

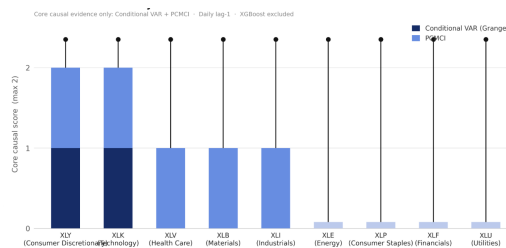


Figure 9: Which Sectors Causally Lead the GSPC

Overall, the industry analysis suggests that sector ETFs and the S&P 500 have a bidirectional lead-lag relationship, but Consumer Discretionary is the strongest and most consistent sector leader. We interpret this as evidence of predictive temporal precedence rather than definitive structural causation.

4 Tech Dominance

4.1 Introduction

Did tech stocks actually cause the S&P 500 to shift over the last 25 years, or were both tech and the broader market simply responding to the same hidden forces at the same time? This section aims to address and answer this question. This distinction cannot be resolved by correlation or regression alone and requires the tools of causal discovery.

To understand why this question matters, we looked into the historical trajectory of the S&P 500 top-10 holdings. In the early 2000s through 2009, spanning the dot-com era through the Global Financial Crisis (GFC), the top 10 holdings were dominated by companies like GE, Exxon, and Pfizer. Tech was present but not dominant. Starting around 2009 through 2017, the recovery era, Apple took rank 1 and held it for 13 consecutive years. From 2017 onward, the FAANG companies, NVIDIA, and Microsoft collectively came to comprise roughly 60% of the entire top 10 weight. By 2025, NVIDIA held rank 1 for the first time ever with a record index weight of 7.98%. What caused this shift?

A naive interpretation would be that tech stocks drove the market upward, but this conflates association with causation. A structurally different explanation is that hidden macro forces, such as interest rate policy, AI investment cycles, and index fund mechanics, repeatedly rewarded the same companies over 25 years, causing both tech returns and the broader market to rise simultaneously without one directly causing the other. This section tests these two competing explanations directly.

4.2 Preprocessing

The preprocessing pipeline to answer this question began with 26 annual CSV files (2000.csv through 2025.csv), with each containing the full S&P 500 constituent list for that year with three columns: Company, Ticker, and Weight & ordered by index weight from largest to smallest. These files were loaded into memory and, for each year, the ten highest weighted constituents were extracted by selecting the rows with the largest weight values, which produced a compact wide format table of 26 rows (one per year) with 30 columns capturing each top ten company's name, ticker, and weight. That wide table was then saved as "top10.csv" and subsequently reshaped into a long format structure of 260 rows, where every row represented a single stock's rank, ticker, company name, and weight for a given year. To identify all unique tickers that had ever appeared in any year's top ten, the ticker columns were flattened and deduplicated, yielding 32 distinct symbols. Monthly historical price data for each of those 32 tickers was then retrieved from Yahoo Finance using the maximum available history at a monthly interval and the resulting date column was standardized to the first day of each month and filtered to retain only the years 2000 through 2025, producing a combined price table of approximately 9,189 rows. This price table was joined to the long format weight table on the shared Ticker and Year keys, so that each monthly price observation inherited the corresponding year's index rank and weight.

The merged result was sorted chronologically by year, rank, and date, and verified to contain no missing weight or company values, arriving at a final dataset of 3,108 rows and 12 columns. In a final enrichment step, each ticker was assigned a binary Sector label, Tech or Non-Tech, based on a predefined set of technology companies present in the dataset, a three period Era label grouping years into the dot-com and Global Financial Crisis era (2000–2008), the recovery period (2009–2016), and the AI and FAANG era (2017–present), and a normalized closing price anchored to each ticker's first observed close multiplied by 100, enabling proportional performance comparisons across stocks with different price scales. The completed dataset was saved as "full_stock_s&p500_weights_monthly.csv" and is available [here](#).

4.3 Data and Variables

To study this question, we constructed five monthly time series spanning 310 observations from February 2000 to December 2025, sourced from Yahoo Finance and the Kaggle S&P 500 holdings dataset. The five variables are `tech_ret`, the weighted-average monthly log-return of tech stocks in the top-10 holdings; `nontech_ret`, the same for non-tech holdings; `sp500_ret`, the equal-weighted average of all top-10 holdings as an overall market proxy; `d_tech_weight`, the first difference

of tech's share of top-10 index weight; and d_top10_conc , the first difference of the Herfindahl concentration index of the top-10 holdings.

4.4 Time Series Exploratory Data Analysis

Before applying any causal methods, we examine the raw price and return behavior of the tech and non tech top 10 holdings across the full 26 year sample. Figure 10 presents two complementary views of the same underlying data.

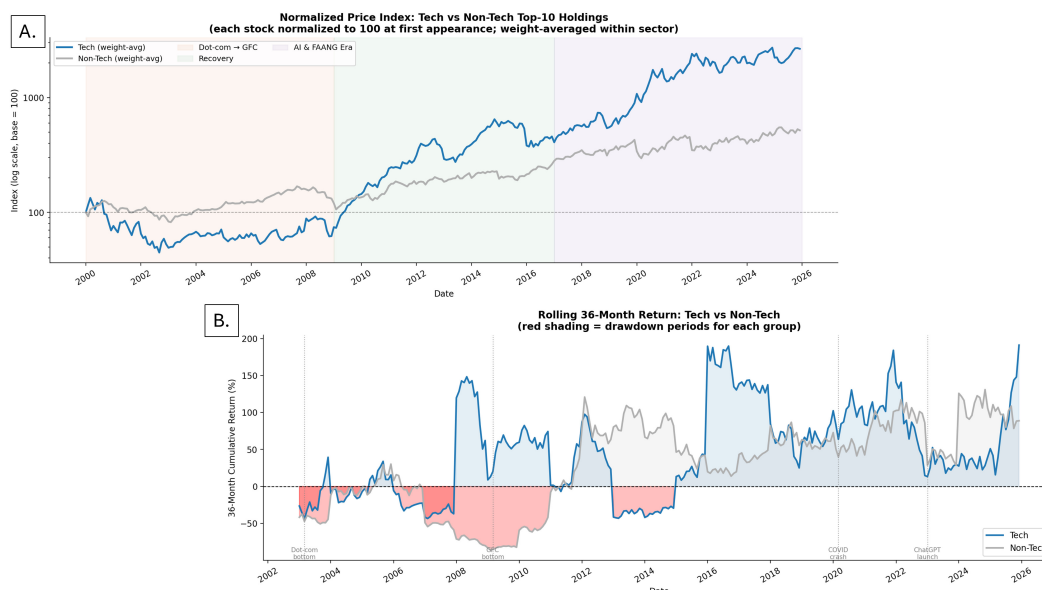


Figure 10: Time series EDA for S&P 500 top-10 holdings (2000–2025). **Panel A** (top): weight-averaged normalized price index for tech and non-tech holdings, each stock anchored to 100 at first appearance, plotted on a log scale with era shading. **Panel B** (bottom): rolling 36-month cumulative return for each sector, with red shading marking drawdown periods and dotted vertical lines at key macro events.

Panel A traces the long-run cumulative price trajectory of each sector on a log scale, allowing structural breaks and sustained divergences to appear visually rather than being obscured by compounding. It shows three structurally distinct phases separated by the era shading. During the dot-com and GFC era (2000 – 2008), tech collapsed to roughly 40 - 50 on the normalized index, losing more than half its value from the 2000 peak while non-tech held close to 100 throughout. This immediate divergence establishes that the two sectors do not move in lockstep even under severe market stress as they share draw downs in the aggregate but absorb losses at very different magnitudes. The recovery era (2009 – 2016) saw both sectors rise together off their GFC lows, with tech overtaking non-tech by around 2010 and accelerating through 2014 before a brief convergence. The most important feature of the plot is the AI and FAANG era (2017 – 2025) as on a log scale, tech compounds to approximately 3,000 (a $30\times$ return from baseline) by end of sample while non-tech reaches only 400 (still a $4\times$ return from baseline), a factor of six divergence that accumulates without any sustained reversal.

Panel B zooms into the medium-run momentum structure by plotting rolling 36-month cumulative returns, annotated with drawdown periods and key macro events. This panel complements Panel A by isolating the medium run momentum dynamics rather than the long run level drift. The most diagnostic feature is the synchronization of drawdowns across both sectors. At the dot-com bottom in 2003, both tech and non-tech simultaneously registered negative 36-month returns, with tech reaching below -50% and non-tech dipping to roughly -40% . At the GFC bottom in 2009, the same pattern repeats where both sectors enter red shaded drawdown territory together, with the timing nearly identical. This co-movement is inconsistent with a story in which tech directly drives the broader market as if tech were the primary cause, one would expect tech to lead and non-tech to follow with a detectable lag. Instead, both sectors fall and recover in near-unison, which is more consistent with a

shared external driver that affects both simultaneously. In the AI and FAANG era, tech’s 36-month cumulative return reaches nearly +200% by end of sample, far outpacing non-tech’s +80–100%, and the gap widened following the ChatGPT launch marker in early 2023. This asymmetry in the magnitude of gains, combined with the symmetry of drawdowns, is the central descriptive pattern that the causal analysis in the following sections is designed to explain. These two panels establish the unconditional picture of how the two sectors moved relative to one another before any conditioning or causal modeling is applied.

4.5 Methods

We applied six methods in sequence, moving from unconditional diagnostics to progressively more assumption-laden causal models. This layered design is intentional as each method is used to check whether the conclusions from the previous one survive stronger assumptions, and agreement across methods with different identifying restrictions substantially strengthens any claim that survives the full sequence.

The first step was stationarity testing. An Augmented Dickey-Fuller (ADF) test at $\alpha = 0.05$ using an OLS-based implementation with up to four lags was tested for all five variables. Table 2 shows the findings present from this test. The three return series `tech_ret`, `nontech_ret`, and `sp500_ret` each rejected the unit root null with ADF statistics of -8.30 , -7.82 , and -7.44 respectively (all with p -values of 0.005), confirming that monthly log-returns are stationary as expected. The two structural series `top10_conc` and `tech_weight` failed to reject the unit root null, with ADF statistics of -1.56 and 0.05 and p -values of 0.20 in both cases, which indicates persistent upward drift with no tendency to revert to a fixed mean. A variable with unit-root behavior has been permanently shifted by forces that accumulate over time rather than reverse, and the nonstationarity of `tech_weight` is direct statistical evidence of a structural regime change independent of any causal graph. Both structural series were first-differenced to produce `d_tech_weight` and `d_top10_conc` before entry into any causal model.

Table 2: Stationarity Testing with Augmented Dickey-Fuller (ADF) Test

Variables	ADF t-stat	p-value	Stationary?
<code>tech_ret</code>	-8.303	0.005	Yes (5%)
<code>nontech_ret</code>	-7.824	0.005	Yes (5%)
<code>sp500_ret</code>	-7.435	0.005	Yes (5%)
<code>top10_conc</code>	-1.563	0.200	No
<code>tech_weight</code>	0.051	0.200	No

The second step examined the raw dependence structure using Pearson correlations (quantifies unconditional linear association), precision-matrix partial correlations (uncovers the residual dependence between two variables after conditioning on all others simultaneously), and Fisher-Z conditional independence tests (provides a formal p -value for the hypothesis that two variables are conditionally independent given a specified conditioning set). These tests serve as direct inputs to the constraint-based causal discovery algorithms that follow.

The third and fourth steps applied constraint-based causal discovery algorithms to the stationary five-variable dataset. The PC algorithm recovers the Markov equivalence class of the underlying causal graph by iteratively testing conditional independence relationships and orienting edges using Meek rules, producing a Completed Partially Directed Acyclic Graph (CPDAG) in which some edges remain undirected due to irreducible equivalence. The Fast Causal Inference (FCI) algorithm extends PC by relaxing the assumption of causal sufficiency. Rather than requiring that all causally relevant variables have been measured, FCI permits the presence of hidden common causes and marks their signature as bidirected edges in a Partial Ancestral Graph (PAG). This distinction matters here because macro forces such as interest rate policy and institutional fund flows are plausibly relevant confounders that are not included in our five-variable system.

The fifth step was pairwise Granger causality. For each ordered pair of variables, we tested whether lagged values of the first variable improve predictions of the second beyond the second variable’s own history, using an OLS F-test at lags one through three. Granger causality is a predictive concept rather than a structural one, but its directional asymmetry (which variables contain information about the future of which other variables) provides a useful complement to the cross-sectional orientation tests

in PC and FCI. Granger causality also has a known vulnerability to false positives when variables share autocorrelation structure, which motivates the sixth step.

The sixth and final step applied PCMCI, a time-series aware extension of PC developed specifically to address the autocorrelation problem. PCMCI first identifies each variable’s own lagged parents via a momentary conditional independence (MCI) test, conditions on those parents before testing cross variable links, and thereby removes spurious associations driven by shared serial structure rather than genuine cross variable predictability. Only links that survive this stricter conditioning are retained. A finding that appears in standard Granger tests but disappears under PCMCI is likely an autocorrelation artifact and a finding that survives both is substantially more credible. The agreement between Granger and PCMCI results is therefore used as a consistency check across the two time series methods, complementing the agreement between PC and FCI on the cross-sectional side.

4.6 Causal Discovery Results: S&P 500 Top-10 Holdings, Tech vs. Non-Tech (2000–2025)

The analysis proceeds in three stages. We first examine the raw dependence structure through Pearson correlation and Granger causality matrices, then trace how Granger significance evolves across lags, and finally interpret the four causal graphs recovered by the constraint-based and time-series causal methods.

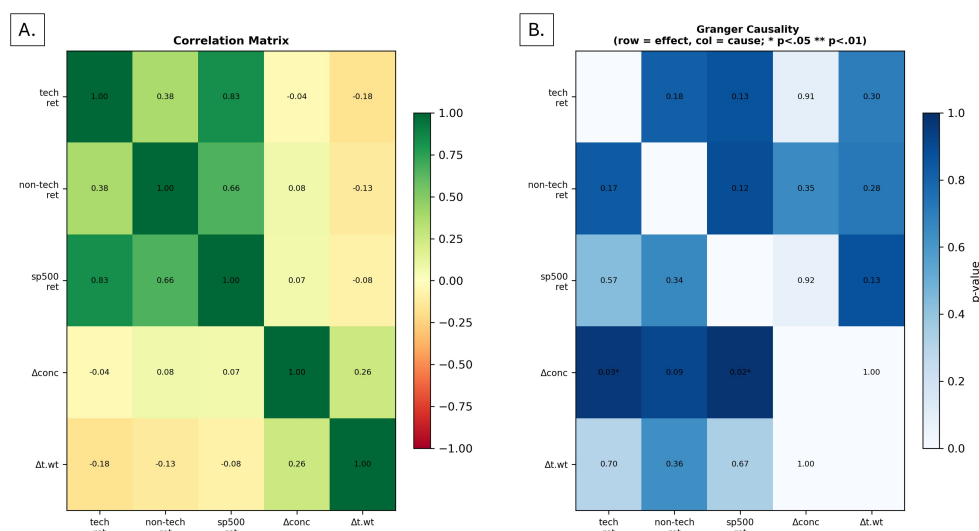


Figure 11: Pairwise dependence structure across the five stationary variables. **Panel A** (left): Pearson correlation matrix; color scale runs from red (-1) through yellow (0) to green ($+1$). **Panel B** (right): pairwise Granger causality p -value matrix at lag 3, where rows index the effect variable and columns the cause; darker blue cells indicate stronger Granger evidence ($*p < .05$, $**p < .01$).

The correlation structure in Figure 11A reveals three distinct clusters. The strongest relationship in the dataset is between `tech_ret` and `sp500_ret` ($r = 0.83$), which is expected given that tech stocks constitute a substantial and growing share of the index proxy. The relationship between `nontech_ret` and `sp500_ret` is moderately strong ($r = 0.66$), while the two sector return series themselves share only a modest correlation ($r = 0.38$). These three return variables form a coherent cluster that moves largely independently of the two structural variables. The two structural variables, `d_tech_weight` and `d_top10_conc`, are weakly but positively correlated with each other ($r = 0.26$), and each shows only minimal marginal correlation with the return series where `d_tech_weight` carries a small negative association with all three return variables (ranging from -0.18 to -0.08), while `d_top10_conc` is near zero across all return variables ($|r| < 0.08$). The near-zero raw correlations between returns and structural changes are the first signal that any causal relationships operating through concentration are likely lagged rather than contemporaneous, which the Granger and PCMCI analyses confirm.

The Granger matrix in Figure 11B is notable almost entirely for what it does not find. Of the twenty directed pairs tested at lag 3, only two clear the $\alpha = 0.05$ threshold: $\text{tech_ret} \rightarrow \text{d_top10_conc}$ ($p = 0.035$) and $\text{sp500_ret} \rightarrow \text{d_top10_conc}$ ($p = 0.017$). Every return-to-return pair fails to achieve significance, meaning past tech returns carry no predictive content for future non-tech returns, and vice versa. No structural variable Granger-causes any return variable at any tested lag, confirming that causality is strictly one-directional where returns drive structural change, but structural change does not feed back to drive future returns. The asymmetry is clean and consistent across all tested pairs.

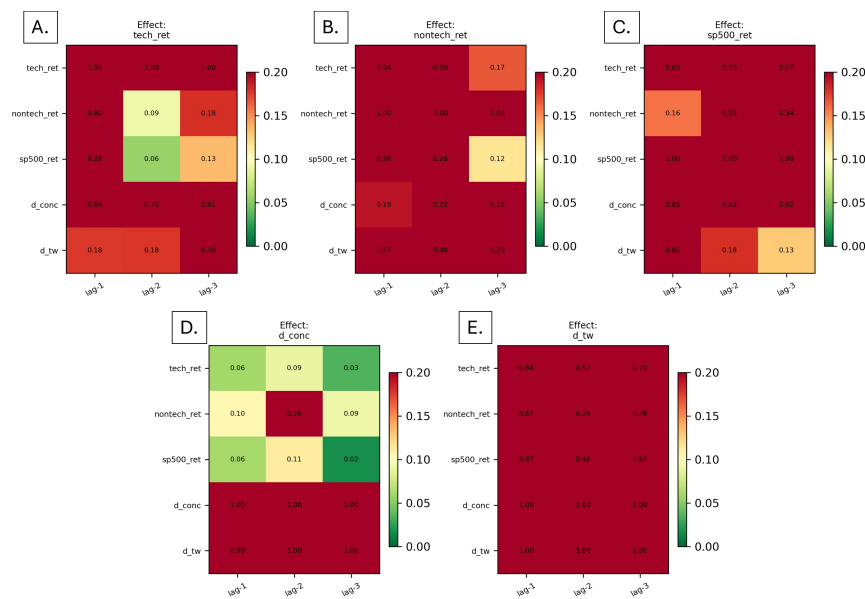


Figure 12: Granger causality p -values broken out by lag for each effect variable. Each heatmap shows the p -value of every potential cause (rows) at lags 1 through 3 (columns) for a single target variable; the color scale runs from green ($p \approx 0$, strong evidence) to red ($p \approx 0.2$, no evidence). **Panel A:** causes of tech_ret . **Panel B:** causes of nontech_ret . **Panel C:** causes of sp500_ret . **Panel D:** causes of d_conc . **Panel E:** causes of d_tw .

In Figure 12A-C, none of the return variable panels produce a single significant cell across all lags and all potential causes. Panel A shows that no variable, including sp500_ret and nontech_ret , reaches $p < 0.05$ for any lag when predicting tech_ret . The closest candidate is sp500_ret at lag 2 ($p = 0.056$), which falls just short of significance. Panels B and C show similarly blank landscapes for nontech_ret and sp500_ret respectively. The return series are not predictable from each other's past values at any of the three lags tested, confirming that the sectors do not drive one another in the Granger sense.

Figure 12D is the only panel with substantive Granger signal. Both tech_ret and sp500_ret show a consistent and strengthening pattern across lags: their p -values begin near 0.06 at lag 1, remain marginal at lag 2 (approximately 0.09 and 0.11 respectively), and cross the significance threshold at lag 3 ($p = 0.035$ and $p = 0.017$). The progressive improvement in p -values from lag 1 to lag 3 is consistent with a slow-moving mechanical relationship: strong price appreciation accumulates within the cap-weighting mechanism over approximately three months before manifesting as a measurable increase in the Herfindahl concentration index of the top-10. The two other potential causes, nontech_ret and d_tech_weight , remain non-significant across all lags.

For Figure 12E, no cause achieves significance for predicting changes in tech's weight share at any lag. Even tech_ret , which does predict concentration changes, does not survive as a predictor of d_tech_weight after conditioning on own lags. This distinction between the two structural variables is informative: concentration is a function of the squared weights of all top-10 stocks and therefore amplifies the effect of extreme outperformers, while tech weight share is a simpler ratio that may be masked by the simultaneous movements of non-tech holdings.

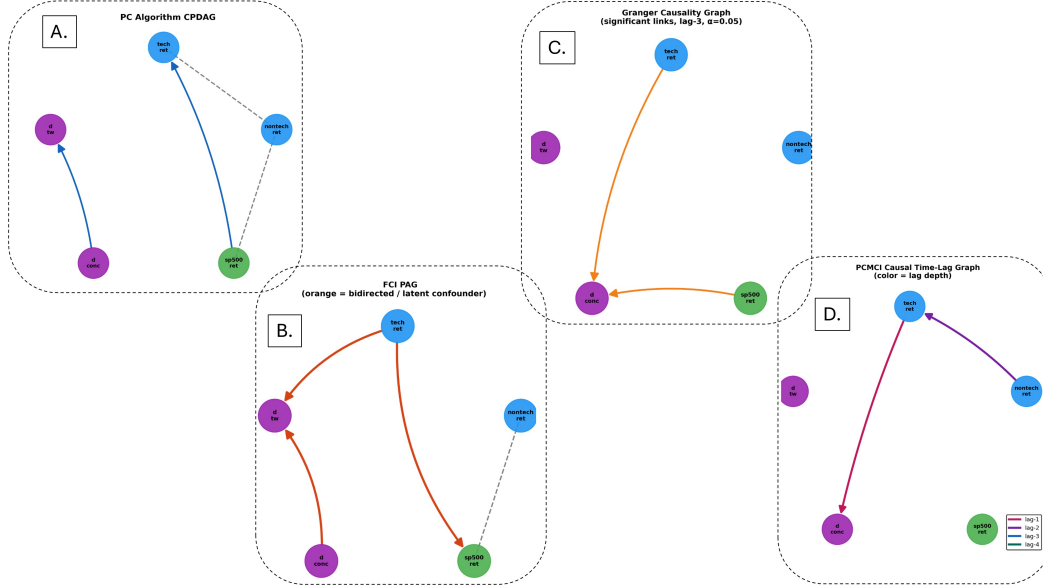


Figure 13: Causal structure graphs recovered by four methods over 310 monthly observations (February 2000 – December 2025). **Panel A:** PC Algorithm CPDAG; blue directed edges are oriented, gray dashed edges remain undirected due to Markov equivalence. **Panel B:** FCI Partial Ancestral Graph; orange bidirected edges indicate a latent common cause, gray dashed edges are undirected. **Panel C:** Granger causality directed graph retaining only edges significant at $\alpha = 0.05$ (lag 3). **Panel D:** PCMCI time-lag causal graph; edge color encodes lag depth (lag-1 through lag-4).

The PC algorithm recovers the minimal causal skeleton consistent with the conditional independence structure of all five variables, then orients as many edges as possible without making assumptions the data cannot support. The two oriented edges tell the primary story. First, $\text{sp500_ret} \rightarrow \text{tech_ret}$ indicates that broad market movements have a directional influence on the tech sector return rather than the reverse, directly contradicting the naive interpretation that tech drives the overall market. Second, $\text{d_conc} \rightarrow \text{d_tw}$ indicates that changes in overall market concentration precede changes in tech's specific weight share. The undirected edges between the three return variables reflect genuine Markov equivalence where conditional independence tests alone cannot orient those edges, which is precisely why the time-series methods in Panels C and D are necessary.

FCI extends PC by relaxing the assumption of causal sufficiency, allowing for the possibility that unmeasured variables are driving observed relationships. Where PC draws a direct arrow, FCI asks whether that arrow could instead reflect shared exposure to a hidden confounder. The answer here is unambiguous: the edges between tech_ret and sp500_ret , between tech_ret and d_tw , and between d_conc and d_tw all become bidirected in the PAG, indicating that latent macro forces are the more parsimonious explanation for those associations. The only surviving directed edge is $\text{nontech_ret} \leftarrow \text{tech_ret}$, recovered through orientation rule R1 after d_tw is identified as a collider. The FCI result is the most epistemically honest representation of the data: without measuring the hidden confounders, most likely interest rate policy, institutional fund flows, and AI investment cycles, we cannot claim clean direct causation between tech returns and the broader index, and any model that does so is imposing an assumption the data does not support.

The Granger graph retains only the edges that survive the OLS F -test at $\alpha = 0.05$ with lag 3. Two arrows remain, both pointing into d_conc : one from tech_ret ($p = 0.035$) and one from sp500_ret ($p = 0.017$). No return-to-return edges survive, and nothing flows out of either structural variable toward any return series. The graph's sparsity is itself a finding. The S&P 500 becomes more concentrated as a lagged consequence of strong performance by tech and the broader market, but that increased concentration does not in turn cause future returns to rise or fall. Under cap-weighted indexing, outperformance mechanically increases a company's share of the index; what Granger causality recovers here is precisely that mechanical feedback loop, operating at approximately a three-month horizon.

PCMCI is the most demanding of the four methods because it conditions on each variable's own lagged parents before testing cross-variable links, stripping out spurious associations driven by shared autocorrelation rather than genuine cross-variable predictive content. After this stricter filtering, only two links survive. The first is $\text{nontech_ret}(t-2) \rightarrow \text{tech_ret}$, with a negative partial correlation of -0.13 ($p = 0.023$), consistent with a mild sector rotation dynamic at a two-month lag: periods of strong non-tech performance are followed by marginal underperformance in tech at a two-month horizon. The second link is $\text{tech_ret}(t-1) \rightarrow \text{d_conc}$, with a positive partial correlation of $+0.12$ ($p = 0.031$). This is the single most credible finding in the entire analysis: it survived conditioning under every method applied, it is consistent with the Granger results at lag 3, and it has a direct mechanical interpretation under cap-weighted indexing. Tech price appreciation in month $t-1$ translates into a larger market-cap weight for tech stocks, which increases the Herfindahl concentration of the top-10 one month later. The arrow runs from performance to structure, not the reverse.

Figure 13 shows these 4 causality graphs.

4.7 Results and Interpretation

This section set out to answer one question: does tech drive the market, or does something else drive both? Across six independent methods, the answer is consistent. There is no direct causal link between tech and non-tech returns. The sectors do not drive each other. Granger causality, FCI, and PCMCI all agree on this conclusion, and the agreement across methods with different assumptions strengthens the claim considerably.

What Granger does consistently show is that tech returns and S&P 500 returns both Granger-cause changes in top-10 concentration with approximately a three-month lag. This is an important asymmetry: returns drive structural change, but structural change does not drive returns. Concentration is a consequence of performance, not a driver of it. Under cap-weighted indexing, strong performance mechanically increases a company's share of the index, which is precisely the relationship the data recovers.

FCI goes further by identifying the mechanism behind this pattern. The bidirected edges between tech returns and S&P 500 returns in the Partial Ancestral Graph confirm that hidden macro forces are driving both simultaneously. What looked like a direct causal link in the PC graph is better explained by shared exposure to unmeasured forces, most likely interest rate policy, AI investment cycles, and institutional fund flows, none of which are directly observed in our dataset. This is the most honest representation of what the data can tell us.

The structural variables provide the final piece of the picture. Both tech weight and market concentration failed the ADF stationarity test with p-values of 0.20, exhibiting persistent upward drift with no tendency to revert. A variable with unit-root behavior is one that has been permanently shifted by forces that do not reverse. Tech's share of the S&P 500 index has drifted upward for 25 years without mean reversion, which is direct statistical evidence that the shift is structural and permanent rather than cyclical.

The takeaway is that tech dominance in the S&P 500 is not tech returns causing the market to move. It is a structural regime change driven by hidden macro forces, confirmed by latent confounders in FCI and the nonstationarity of tech's index weight. The market did not shift because tech drove it. It shifted because 25 years of macro forces repeatedly rewarded the same companies, and cap-weighted indexing mechanically recorded the result.

5 Conclusion

Across three sections and six causal methods, a consistent and coherent picture of the S&P 500's causal structure was formed. The findings do not contradict each other. They reinforce each other at different scales, building a layered account of how the market actually works.

Macro Level

At the macro level, two clean causal pathways were recovered from 25 years of data using the PC algorithm, CD-NOD, and backdoor-adjusted effect estimation. The first is a direct market driver channel: VIX and Oil directly cause S&P 500 returns after controlling for all other variables. The second is the Fed reaction function: CPI, unemployment, and VIX directly drive Fed rate decisions, consistent with the Taylor Rule and dual mandate. Notably, CPI's direct effect on stock returns is near zero after backdoor adjustment, meaning inflation moves markets only by triggering rate changes, not by directly affecting valuations.

CD-NOD revealed that while the causal architecture is stable across the full sample, the dominant channels shift across regimes. M2 drove the system pre-GFC, oil and bond yields took over during ZIRP as QE broke monetary transmission, and CPI with labor market conditions became the primary Fed drivers post-COVID. These regime shifts match official Federal Reserve policy explanations closely, providing strong external validation of the approach.

Sector Level

At the sector level, causality flows in both directions between sectors and the market, but not symmetrically. At daily resolution, almost every sector Granger-causes the S&P 500. Narrowing to where both PCMC and Conditional VAR Granger agree after conditioning on all other sectors simultaneously, only Consumer Discretionary and Technology survive as causal leaders. Consumer Discretionary is the single clearest first mover across every method and time resolution in the analysis. This reflects its business-cycle sensitivity, responding earliest to changes in consumer confidence and credit conditions. This result represents predictive temporal precedence rather than structural causation, as a shared macro shock could reach Consumer Discretionary first and then propagate to the rest of the market.

Index Composition Level

At the index composition level, six methods converged on the same answer. No direct causal link exists between tech and non-tech returns. FCI found bidirected edges between tech and S&P 500 returns, confirming that hidden macro forces drive both simultaneously. The single link that survived every method was tech returns causing increases in top-10 concentration with approximately a one-month lag, consistent with cap-weighted indexing mechanics. The structural variables, tech weight and market concentration, both failed the ADF stationarity test, providing direct statistical evidence of a permanent regime change rather than a cyclical pattern.

Unified Interpretation

The thread connecting all three sections is this that macro forces set the causal regime, consumer discretionary and technology move first within it. That performance compounds over time under cap-weighted indexing, mechanically concentrating the index toward tech. The S&P 500 did not shift because tech drove it. It shifted because 25 years of macro forces repeatedly rewarded the same companies, and the index construction rules did the rest.

This consistency across methods and scales is the core contribution of this project. Each of the six methods used is independent in the sense that it makes different assumptions, uses different test statistics, and targets different aspects of causal structure. The fact that they converge on the same picture is not a coincidence. It is evidence. The causal structure of the S&P 500 is real, it is measurable, and it is stable enough to survive scrutiny from multiple independent angles.

Our code is available [here](#).

6 Future Directions and Limitations

Limitations

There are a few limitations that are present in our study. First, from the data standpoint, the data used for the 3 sub-studies only contain 9 sectors, excluding real estate, communication services, and international markets. As these sectors were not included, it is unknown whether these sectors also play a role in the causal relationships studies. There is possibility that they contribute to the causal relationship driven by macroeconomic factors, industries, or play a role in the rise of the tech industry. Also, in the macro and tech sections, the monthly data used is limited, having low numbers of observations, which means it is harder to detect causal relationships.

Second, when looking at the methods used in the analysis across all three sections, Granger causality and PCMCI are assuming linear relationships. This omits possible nonlinear causal relationships in the study. Across the three sections, the data has been assumed to be sufficient, however there may be confounding variables lurking around that also have a role in the causal relationships studied. Not only that, PCMCI and VAR causality tests are sensitive to the chosen significance level. So this analysis is valid under the current conditions set, if the significance level is changed, then results may change as well.

Future Directions

One important future improvement is to extend the data and analysis to including inter-day data as well, with data at various selected time-intervals. This may give us better insight to the causal relationships with real-time and constant changing data.

It will also be interesting to see the connection between the three different approaches to the stock market that has been taken. For example, it will be interesting to combine the macro and sector level data into one system and create a causal infrastructure within this proposed system to observe relationships. It will then be possible to see if the sectors or market drive macroeconomic indicators, or whether it is the other way around. Building upon this, can then analyze specific industries and their relationship to macroeconomic indicators.

It will also be interesting to apply the backdoor criterion to the sector-to-market edges, which was done in the macro section. This will allow us to estimate actual causal effect sizes than just binary support scores.

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7 Appendix

7.1 Figures

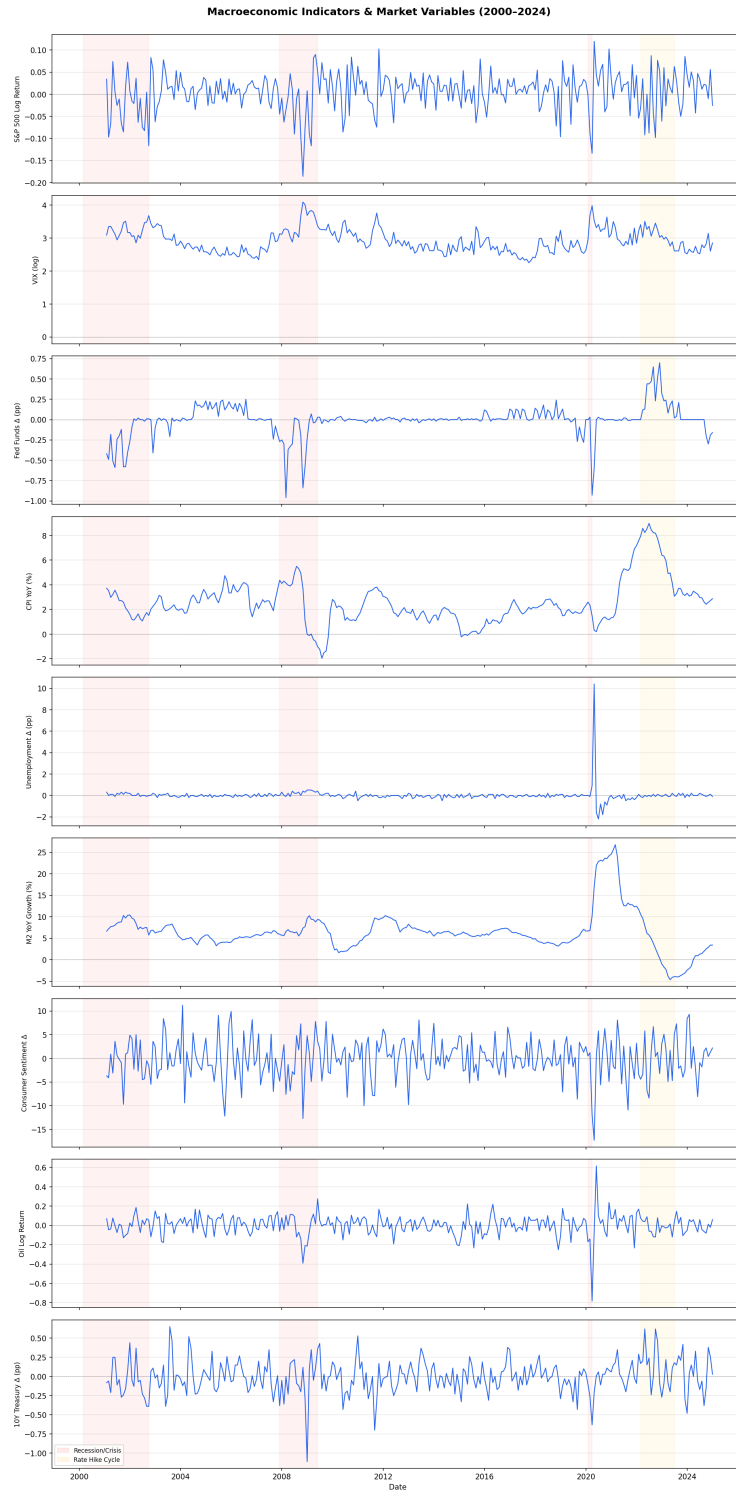


Figure 14: All macroeconomic variables plotted as time series

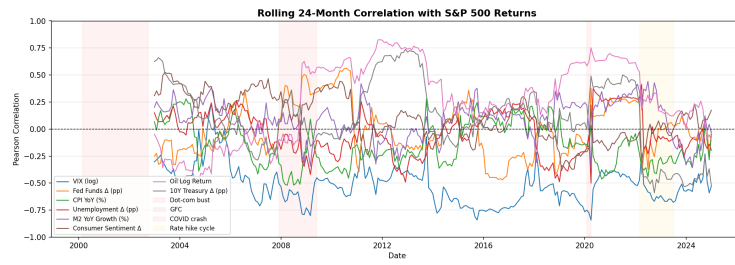


Figure 15: Rolling correlation between all variables and the S&P 500 return